Proposed Title: Additive Manufacturing Based Artificial Fog Collection System to Increase Water Condensation Efficiency – A Multibioinspried Approach

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Introduction and Overview

The goal of the proposed research project is aiming to improve water condensation efficiency, by designing an additive manufacturing-based artificial fog collection system which takes a multibioinspired approach, mimicking the functioning of natural mechanisms seen in things like spider webs, rice leaves, cactus spines, and Nepenthes alata. This project aims to meet the critical need for sustainable water collection technologies, especially in arid and semi-arid regions where fog, a significant yet underutilized water source, holds immense potential. With the global

population growing, water resources are being depleted to levels that may never be replenished. Remarkably, 10% of Earth's freshwater is stored in the atmosphere as fog, representing a vast, untapped resource [1]. The innovative 3D printing technology will be adopted to allow for the creation of surfaces with micro- and nanoscale patterns that are reminiscent of these naturally occurring water-harvesting structures, which will enhance the capture and condensation of atmospheric water in fog-rich situations by improving surface morphology, wettability, and structural patterns. This bio-inspired design, when coupled with additive manufacturing, will provide an economical and scalable answer to the problem of water shortage, which is particularly acute in dry locations.

Prior investigations have suggested several structures and approaches for the collection of fog water. For instance, Liu et al. presented a hybrid fog collector with cactus-inspired spines bearing longitudinal ridges and peristome-inspired bottom channels with curved inclined arc-pitted grooves (C-IAPGs) and found that printed spines with four longitudinal ridges capture more fog and the bottom channel with C-IAPGs efficiently transports water droplets into the reservoir [1]. In another study, Li and colleagues introduced a novel approach in which they developed a cactus-inspired surface adorned with several directional artificial spines coated with nanoscale hydrophobic material on the surface of the 3D-printed spines to enhance the efficiency of water collection and conveyance. The results indicated that the hexagonally distributed clusters improve the circulation of moisture around 3D-printed spines with a 10° tip angle and hydrophobic coating show the greatest increase in weight [2]. Moreover, Jamil et al. present a bioinspired slippery surface comprising of candle soot particles that use their roughness to promote the development of condensed water droplets. Combined, the Namib desert beetles-inspired convex topography, cacti-inspired asymmetric slope, and Nepenthes pitcher plant-inspired slippery surface of soot particles provide a high droplet nucleation density, rapid droplet coalescence, and effortless droplet removal from the surface [3].

Although previous studies on fog collection have shown successful outcomes in terms of their specific structure and methods, the mass flow rate of water condensation varies among the individual studies. The condensation of fog water remains an issue in real-world applications. However, the integration of individual structures with advanced additive manufacturing techniques can help alleviate this problem.

We **propose** to address these knowledge gaps via **Additive Manufacturing Based Artificial Fog Collection System to Increase Water Condensation Efficiency – A Multibioinspried Approach** with the following **objectives**:

(1) To improve fog collecting, additive manufacturing to create micro- and nanoscale surface patterns inspired by cactus spines, spider silk, rice leaves, and Nepenthes alata's water-harvesting properties.
(2) Optimize bioinspired designs' wettability, surface texture, and structural orientation to enhance water condensation, capture, and droplet movement.
(3) Measure water collection rates, droplet coalescence, and transport dynamics to evaluate the bioinspired fog collection surfaces in controlled fog chamber conditions.
(4) Create a scalable additive manufacturing technique (μ-CLIP or FDM 3D Printing) for fog collectors for real-world deployment in fog-prone and water-scarce places to address global water issues.

Intellectual merit

This research leverages the principles of biomimicry and advanced 3D printing technologies to create an innovative fog collection system that replicates nature's most effective water harvesting strategies. By integrating the water collection mechanisms of cactus spines, which promote directional water transport through optimized surface wettability gradients, spider silk with its alternating hydrophilic and hydrophobic regions to enhance droplet adhesion and movement, and rice leaf microchannels, which use low adhesion and micro-scale capillary action for rapid water transport, the system significantly improves fog collection efficiency. The inclusion of C-IAPG channels further enhances droplet transport dynamics by ensuring rapid, controlled water movement through asymmetrical grooves. The system also features high wettability contrast between hydrophobic and hydrophilic zones, improving droplet capture and transport, while low adhesion surfaces prevent water retention and enable faster collection. Additionally, superhydrophobic coatings, such as those derived from candle soot, enhance water condensation and reduce surface energy for more efficient droplet sliding. This interdisciplinary project bridges materials science, biomimetic engineering, and environmental science, advancing the frontiers of sustainable water collection technology. By incorporating features like controlled wettability, optimized surface adhesion, and advanced microchannel designs, this research fosters innovation in water resource management and addresses the growing global challenge of water scarcity.

Broader Impacts

Specifically, the Additive Manufacturing-Based Artificial Fog Collection System has significant implications for tackling worldwide water shortage and advancing sustainable resource management. Utilizing bioinspired designs and cutting-edge manufacturing technology, this system will offer a scalable and energy-efficient alternative for collecting water in areas prone to fog and aridity, where conventional water supplies are scarce. This technology's capacity to enhance water condensation efficiency will greatly enhance the availability of clean water for rural and underprivileged areas, therefore promoting social and economic progress. Furthermore, the system's sustainable design is in line with worldwide initiatives to address climate change by decreasing reliance on energy-intensive desalination and groundwater extraction techniques. The use of additive manufacturing not only provides environmental and socioeconomic advantages

but also stimulates innovation in material science and engineering, therefore fostering the advancement of more sophisticated and environmentally friendly water-harvesting devices.

Motivation and Background

Presently, the severity of water scarcity has escalated due to the inadequacy of the current water supply in meeting the growing worldwide population's water demand [4]. In response to this issue, numerous initiatives have been undertaken to acquire freshwater, including fog harvesting [5], desalination [6], membrane filtration [7], and solar-powered water production [8]. An appealing technique for capturing atmospheric water, fog collection, can be applied even in arid (e.g., deserts) and contaminated water situations. Furthermore, in comparison to other techniques, fog collection offers the benefit of not only being unrestricted by geography yet also enabling quick on-site production [9]. Through millions of years of development, numerous living organisms have developed distinct mechanisms for collecting fog in order to survive in arid and semi-arid environments [10]. For example, Namibia Desert beetles and spider silks are highly efficient at collecting water from fog [11]. Moreover, several species of the Cactaceae family can thrive in highly dry conditions by the development of spines that collect fog from many directions [12].

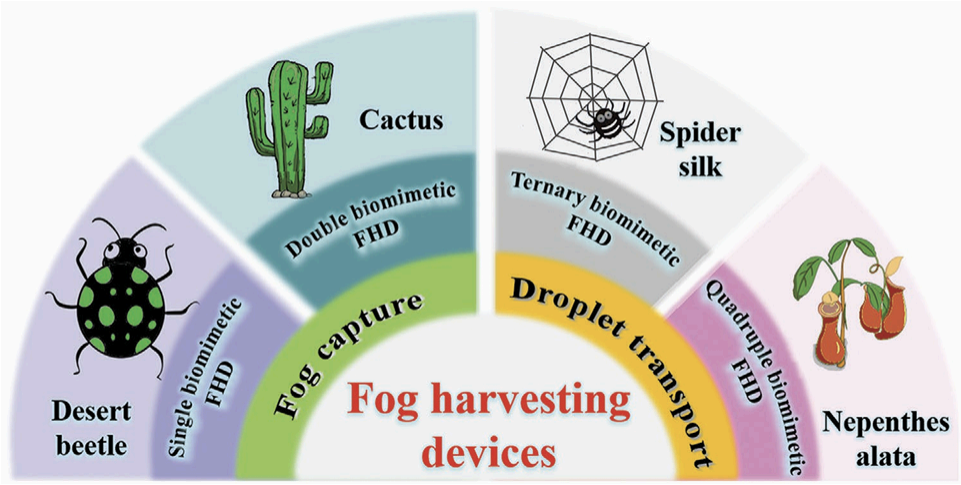

Figure 1. The state-of-the-art development of FHDs: introduction of typical creatures, fog harvesting mechanisms, and devices fabrication inspired from single creature to multiple creatures [11].

In contrast, previously documented bioinspired fog collecting materials and structures are frequently produced using conventional techniques, such as replica modeling [13], electrochemical erosion of metal wire [14], and hydrothermal synthesis [15]. Nevertheless, the task of creating complex 3D buildings using traditional techniques remains difficult. Conversely, 3D printing, also known as additive manufacturing (AM), has quickly developed as a means of producing bioinspired structures, a task that is tough for conventional techniques [1], [16]. For the fabrication of artificial fog collectors, AM has significant benefits, such as the ability to precisely regulate surface textures at the micro- and nanoscale, which enhances the effectiveness of water condensation by fabricating of intricate architectures that enhance water absorption, while simultaneously reducing material wastage by its sequential layering technique. The rapid

prototyping capacity of this technology enables swift testing and optimization of designs. Moreover, AM facilitates scalability and customization, allowing for the fabrication of structure in different dimensions and arrangements which makes AM a highly efficient, adaptable, and environmentally friendly manufacturing technique.

Objectives, Hypotheses, and Research Plan

The aim of this study is to create an Artificial Fog Collection System using Additive Manufacturing methodology. This system will improve the efficiency of water condensation by employing a multibioinspired strategy inspired by the water-harvesting abilities of cactus spines, spider silk, rice leaves, and Nepenthes alata (pitcher plant). The hypothesis posits that the integration of micro- and nanoscale surface structures, influenced by the directional water movement of cactus spines, the hydrophilic-hydrophobic patterning of spider silk, the superhydrophilicity of rice leaves, and the slippery surfaces of Nepenthes, will greatly enhance the efficiency of water capture and transport in fog collectors. The study proposal entails the development and 3D printing of diverse surface patterns that imitate natural structures. Subsequently, experimental testing and numerical analysis will be conducted in controlled fog chamber settings to quantify water collection rates and enhance the designs for practical implementation in areas prone to fog. Project Schedule and Milestones

The research milestones are divided into the following tasks and described accordingly, followed by a research task schedule in table 1

Table 1: Research Task Schedule

Week	Objectives	Tasks and Description
Week 1	Literature Review	Objective: Build foundational knowledge on bioinspired designs and technologies. Tasks: 1. Study cactus spines, spider silk properties, rice leaf microchannels, and C-IAPG grooves. 2. Review superhydrophobic coatings and advanced 3D printing techniques.
Week 2	Literature Review and Design Planning	Objective: Finalize understanding and begin planning the integrated design. Tasks: 1. Summarize the key findings from the literature. 2.Plan how to integrate all 4 designs into a single design using Solidworks. 3. Identify suitable materials for prototyping
Week 3	Design Phase	Objective: Create the integrated prototype design. Tasks: 1. Use Solidworks to model the combined design incorporating: (i) Cactus-inspired spines. (ii) Spider silk-inspired gradient surfaces. (iii) Rice leaf microchannels. (iv) C-IAPG channels. 2. Prepare design for simulation.
Week 4	Simulation Phase	Objective: Simulate the prototype's performance and optimize the design. Tasks: 1.Import design into COMSOL Multiphysics or similar software. 2. Run simulations on: (i) Fog collection efficiency. (ii) Water transportation dynamics. (iii) Structural integrity. 3. Refine designs based on results.
Week 5	Prototyping Phase	Objective: Fabricate the prototype and apply surface coatings. Tasks: 1. Select appropriate 3D printing materials compatible with coatings. 2. Optimize 3D printing parameters for precision. 3. Print and assemble the prototype. 4. Apply superhydrophobic coatings to necessary surfaces.
Week 6	Initial Testing and Optimization	Objective: Test the prototype and make necessary adjustments. Tasks: 1. Set up controlled experiments to measure: (i) Fog collection efficiency. (ii) Droplet movement and water transport. 3. Analyze data and identify areas for improvement. 4. Modify design and reprint components if needed.
Week 7	Final Testing and Scalability Evaluation	Objective: Conduct comprehensive testing and assess scalability for mass production. Tasks: 1. Perform durability and longevity tests under real world conditions. 2. Evaluate mass production feasibility. (i) Cost estimation. (ii) Product time analysis.
Week 8	Reporting amd Presentation	Objective: Compile findings and prepare for dissemination. Tasks: 1. Analyze all collected data and finalize conclusions. 2. Prepare a comprehensive report detailing methodolgy, results, and recommendations.

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